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Tensionable knotless anchors and methods of tissue repair

Abstract

Tensionable, knotless, self-locking surgical constructs and methods for surgical repairs are disclosed. A soft suture anchor and a flexible strand (flexible coupler) are provided integrally, as a one-piece machine taper construct. The repair system can include one or more shuttle/pull devices. The suture comes out of the cannulated sheath and braids to a smaller diameter round suture which is then passed back through the sheath. The braided suture may be furcated or branched to provide a plurality of passing sutures.

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Background/Summary

BACKGROUND
(1) The disclosure relates to the field of surgery and, more specifically, to knotless anchor constructs and associated methods of tissue repairs.
SUMMARY
(2) Reconstruction systems, assemblies and methods for fixation of soft tissue are disclosed.
(3) A tensionable, knotless surgical construct can create a knotless, self-locking, reinforced repair. A tensionable, knotless, self-locking surgical construct is made completely of suture(s) to achieve fixation in bone without a separate anchoring body. The design enables fixation by deployment into bone with suture tail(s) remaining outside the bone for tensioning and/or alternative usage. The knotless surgical construct may be employed in knotless fixation of first tissue to second tissue, for example, fixation of tendon to bone.
(4) Methods of tissue repairs are also disclosed. In an embodiment, a knotless surgical construct provides tissue fixation without any knot formation, by providing an all-suture soft anchor device that does not require a separate anchoring body or similar structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1-5** illustrate schematic views of surgical constructs.

(2) FIG. **6** illustrates a schematic repair with surgical construct of FIG. **1**.

DETAILED DESCRIPTION

(3) A tensionable, knotless, self-locking surgical construct can create a knotless, reinforced repair.

(4) A soft tissue repair system includes a tensionable, knotless, self-locking surgical construct with a fixation device and a flexible strand (flexible coupler) provided as a one-piece machine taper construct. The repair system can include one or more shuttle/pull devices.

(5) In an embodiment, a soft suture anchor includes a single suture knotless anchor, wherein a soft anchor sheath is provided integral with the repair suture(s) (a "single suture knotless FiberTak® soft anchor"). The suture comes out of the cannulated sheath and braids to a smaller diameter round suture which is then passed back through the sheath. The braided suture may be furcated/branched/forked (for example, bifurcated or trifurcated, etc.) to provide a plurality of passing sutures.

(6) The construct secures tissue (arthroscopic or endoscopic) with a suture that does not require the surgeon to tie a knot. The construct pulls soft tissue together. The furcation(s) in the suture allow the construct to improve tissue cut-through resistance (by reducing the "cheese wire" effect) which is important in many clinical applications. The knotless surgical construct can be employed in knotless fixation of first tissue to second tissue, for example, fixation of soft tissue to bone.

(7) Methods of knotless tissue repairs are also disclosed. In an embodiment, a surgical construct provides knotless first tissue to second tissue fixation, for example, knotless fixation of soft tissue (ligament, tendon, graft, etc.) to bone without any knot formation and in a simple and fast manner.

(8) The sheath and braided suture can be manufactured from any flexible material, for example, multifilament, braided, knitted, woven suture, or including fibers of ultrahigh molecular weight polyethylene (UHMWPE) or the FiberWire® suture (disclosed in U.S. Pat. No. 6,716,234, the disclosure of which is hereby incorporated by reference in its entirety herein). Surgical constructs can be used with any type of flexible material or suture known in the art. The shuttle/pull device can be a suture passing instrument or a shuttle link such as a FiberLink™ or a Nitinol loop.

(9) Referring now to the drawings, where like elements are designated by like reference numerals, FIGS. **1-5** illustrate exemplary surgical construct **100**, **200**, **300**, **400**, **100***a* (implantable device **100**, **200**, **300**, **400**, **100***a*; surgical assembly **100**, **200**, **300**, **400**, **100***a*; surgical system **100**, **200**, **300**, **400**, **100***a*; suture anchor **100**, **200**, **300**, **400**, **100***a*; tensionable, knotless construct **100**, **200**, **300**, **400**, **100***a*; tensionable, knotless, self-locking, surgical anchor **100**, **200**, **300**, **400**, **100***a*) including exemplary fixation device **10** with exemplary knotless, self-locking, tensionable mechanism **199** within a body of the fixation device **10**. FIG. **6** illustrates an exemplary soft tissue repair **101** with surgical construct **100**.

(10) As detailed below, surgical construct **100**, **200**, **300**, **400**, **100***a* is an implantable device made completely of suture or sutures that achieves fixation in bone without a separate anchoring body. The design enables for fixation by deployment into bone with suture tail or tails remaining outside the bone for tensioning and alternative usage.

(11) Surgical construct **100** of FIG. **1** includes fixation device **10** formed integral (machined integrally) with flexible coupler **20** (flexible strand **20**; suture **20**; repair suture **20**). One or more shuttle/pull devices **40** (suture passing instrument **40**; suture passer **40**; shuttle link **40**; FiberLink™ **40**; nitinol loop **40**) are provided attached to fixation device **10**.

(12) As shown in FIG. **1**, fixation device **10** can be in the form of a soft anchor (soft suture anchor, or all-suture soft knotless anchor) provided with a soft anchor sleeve **10** (sheath or tubular member **10**) and a cannulation **11** (not shown) with two ends **12**, **13**. One of the two ends, for example, end

12 is an open end. The other of the two ends, for example, end **13** terminates in, and continues with, flexible coupler **20**, i.e., flexible coupler comes out of the cannulated sheath **10** (soft anchor sleeve or sheath) and braids to a smaller diameter strand **20** having a single flexible end **21**. In an embodiment, flexible coupler **20** diameter may be the same or smaller suture having a round diameter. In an embodiment, flexible coupler **20** is a single strand of #5 or #2 machine taper suture.
(13) Flexible coupler **20** is passed back through the cannulation **11** of sheath **10** to form at least one flexible, closed, knotless, continuous, adjustable loop **50** having an adjustable perimeter and adjustable length.
(14) FIG. **1** also illustrates one or more shuttle/pull devices **40** extending through the sleeve **10**. If more than one shuttle/pull devices **40** are provided, the shuttle/pull devices can extend through the sheath **10** in similar or different directions and/or orientations and/or locations. Details of an exemplary soft suture anchor with a soft anchor sleeve (sheath or tubular member) and flexible shuttling strands are set forth, for example, in U.S. application Ser. No. 15/998,516 entitled “Methods of Tissue Repairs” filed on Aug. 16, 2018 (issued as U.S. Pat. No. 10,849,734 on Dec. 1, 2020), the disclosure of which is incorporated by reference in its entirety herein. The flexible coupler and the one or more shuttle/pull devices can extend through the sleeve in similar or different directions and/or orientations and/or locations.
(15) Flexible coupler **20** is provided with a single terminal end **21**. The terminal end **21** is a flexible end (free end **21**) that is formed integral with end **13** of the sheath **10**. Flexible coupler **20** comes off of the cannulated sheath **10** and is braided to a round suture with a diameter “d” which may be smaller than diameter “D” of the sheath **10**. The diameter differentiation and furcation of suture occurs within the manufacturing process (i.e., braiding, weaving, sewing, stitching, etc.). In an embodiment, flexible coupler **20** is a one-piece machine taper construct with sheath **10** and has the form of a round single suture strand of #5 or #2 machine taper suture. The sheath **10** acts as the soft sheath. Shuttle suture **40** (shuttle/pull device **40**) is passed through the sheath **10**. Shuttle suture **40** (shuttle/pull device **40**) is attached to the sheath **10** and/or flexible coupler **20** by being spliced, for example, to the flexible coupler.
(16) As detailed below, end **21** can form one or more flexible, closed, knotless, continuous, adjustable loops **50***a*, **50***b*, **50***c* . . . etc. (referred to, for simplicity, as “flexible, closed, knotless, continuous, adjustable loops **50**”) having an adjustable perimeter and adjustable length.
(17) FIG. **1** illustrates one exemplary shuttle/pull device **40** in the form of a suture passing instrument or a suture passer such as FiberLink™ **40** or nitinol loop **40**. However, the disclosure is not limited to this exemplary-only embodiment and contemplates implantable devices provided with a plurality of shuttle/pull devices. If more than one shuttle/pull device **40** is employed, the plurality of shuttle/pull devices can reside within the sheath **10** independent from each other. Suture passing device **40** includes an eyelet/loop **43** for passing the flexible coupler **20**. If a plurality of shuttle/pull devices are employed, each of the plurality of shuttle/pull devices can include a corresponding eyelet/loop **43** for passing the flexible coupler **20**. The shuttle/pull devices may be similar to each other or different from each other.
(18) The flexible coupler **20** can be passed through at least a portion of the body of the fixation device **10**, for example, through a full cannulation of the fixation device, or may enter and/or exit the body of the fixation device at a location other than most distal end and most proximal end of the fixation device, for example, location A in FIG. **1**.
(19) As detailed below, either the device is inserted into bone via self-punching driver or, alternatively, subsequent to the insertion of fixation device **10** of surgical construct **100** into a drilled hole in bone **90**, the flexible coupler **20** and shuttle/pull device **40** are released from the driver, and the driver removed. Free end **21** of flexible coupler **20** is subsequently passed around or through tissue **80** and then through the eyelet/loop **43** of suture passing device **40**. Suture passing device **40** is then pulled, thereby pulling free end **21** of the flexible coupler **20** towards the body of the fixation device, inside of the sheath **10** and then exiting the cannulation **11** of sheath **10** to form

another flexible, closed, knotless, continuous, adjustable loop **50**. Free end **21** forms a splice inside the sheath **10**. The suture end **21** of flexible coupler **20** can then be tensioned and cut.

(20) FIGS. **2-4** illustrate soft suture anchors **200**, **300**, **400** which are about similar to soft suture anchor **100** of FIG. **1** in that they are also one-piece machine taper constructs and they include flexible couplers **20***a* and **20***b* provided with a single terminal end **21** that are formed integral with the sheath **10**, and shuttle/pull device **40** passed through the sheath **10**. However, suture anchor **200** includes flexible couplers **20***a* and **20***b* which are bifurcated from the sheath **10** and provide a plurality of loops.

(21) As in the previous embodiment, each of the legs **20***a*, **20***b* is passed through the cannulated body of the sheath **10** to form first and second flexible, closed, knotless, continuous, adjustable loops **50***a*, **50***b*. Free end **21** of flexible coupler **20** can be subsequently passed around or through the tissue **80** and then through the eyelet/loop **43** of suture passing device **40**. Suture passing device **40** is then pulled, thereby pulling free end **21** of the flexible coupler **20** (together with bifurcated legs **20***a*, **20***b*) towards the body of the fixation device, inside of the sheath **10** and then exiting the sheath **10** to form additional flexible, closed, knotless, continuous, adjustable loops **50***c*, **50***d* (not shown). The suture end **21** of flexible coupler **20** can then be tensioned and cut. Alternatively, with the removal of the suture end **21**, the legs **20***a*, **20***b* can be set free for formation of additional loops (if additional shuttle/pull devices **40** are provided) or employed for additional surgical steps and as necessary.

(22) FIG. **3** illustrates construct **300** which is also a one-piece machine taper construct with flexible sheath **10** terminating into two suture legs **20***a*, **20***b*. One of the two suture legs (for example, suture leg **20***b*) is passed through the sheath **10** to form flexible, closed, knotless, continuous, adjustable loop **50**. Free end **20***b* can be subsequently passed around and/or through the tissue **80** and then through the eyelet/loop **43** of suture passing device **40**. Suture passing device **40** is then pulled, thereby pulling leg **20***b* towards the body of the fixation device, inside of the sheath **10** and then exiting the sheath **10** to form an additional flexible, closed, knotless, continuous, adjustable loop. Suture leg **20***a* can be used for other tissue repairs that could include knotted repairs, such as lateral tissue compression with accompanying interference anchors, for example.

(23) FIG. **4** illustrates construct **400** which is also a one-piece machine taper construct with flexible sheath **10** terminating into two suture legs **20***a*, **20***b*. One of the two suture legs (for example, suture leg **20***b*) is passed through the sheath **10** to form flexible, closed, knotless, continuous, adjustable loop **50**. The other leg (for example, suture leg **20***a*) is passed through the sheath **10** and exits at location B to form another loop **51**, for additional reinforcement of the construct. Free end **20***b* can be subsequently passed around or through the tissue **80** and then through the eyelet/loop **43** of suture passing device **40**. Suture passing device **40** is then pulled, thereby pulling leg **20***b* towards the body of the fixation device, inside of the sheath **10** and then exiting the sheath **10** to form an additional flexible, closed, knotless, continuous, adjustable loop. Suture leg **20***a* can be used for other tissue repairs that can include knotted repairs, lateral tissue compression conducted with accompanying interference anchors, etc.

(24) FIG. **5** illustrates yet another soft suture anchor **100***a* which is about similar to soft suture anchor **100** of FIG. **1** in that it is also a one-piece machine taper construct which includes a flexible coupler **20** provided with a single terminal end **21** which is formed integral with the sheath **10**, and shuttle/pull device **40** passed through the sheath **10**. However, suture anchor **100***a* differs in how coupler **20** pierces and exits the sheath **10**, i.e., it enters the sheath **10** at the same end **13** where it exits to form a first loop or eyelet **50***a*, passes through the cannulation **11** of the sheath **10**, exits at location A at the other end **12**, and passes back through first loop **50** to form another loop **50***b*.

(25) Construct **100***a* of FIG. **5** is a one-piece machine taper construct sheath **10** that contains a proximal eyelet/loop **50***a* and exits at location A to form another loop **50***b*, for additional reinforcement of the construct. Free end **21** can be subsequently passed around or through the tissue **80** and then through the eyelet/loop **43** of suture passing device **40**. Suture passing device **40**

is then pulled, thereby pulling leg **21** towards the body of the fixation device, inside of the sheath **10** and then exiting the sheath **10** to form an additional flexible, closed, knotless, continuous, adjustable loop. Suture leg **21** can be used for other tissue repairs. Fixation device **10** which can be in the form of a soft anchor (soft suture anchor, or all-suture soft knotless anchor) provided with a soft anchor sleeve **10** (sheath or tubular member **10**) and a cannulation **11** (not shown) with two ends **12**, **13**. One of the two ends, for example, end **12** is an open end. The other of the two ends, for example, end **13** terminates in, and continues with, flexible coupler **20**, i.e., flexible coupler comes out of the cannulated sheath **10** (soft anchor sleeve or sheath) and braids to a smaller diameter strand **20** which ends in a single flexible end **21**. In an embodiment, flexible coupler **20** may or may not have a smaller diameter suture having a round diameter. In an embodiment, flexible coupler **20** is a single strand suture.
(26) FIG. **6** illustrates a schematic tissue repair **101** (e.g., tendon or ligament repair) with exemplary surgical construct **100** of FIG. **1**, to secure a first tissue **80** (for example, soft tissue such as tendon **80**) to a second tissue **90** (for example, bone **90**). Once flexible sheath **10** of surgical construct **100** has been inserted and secured within a hole in bone **90**, the flexible end **21** of flexible coupler **20** is passed around or through tissue **80** and then through eyelet **43** of shuttle/pull device **40**. The shuttle/pull device **40** is then pulled out of the fixation device **10** and out of the surgical site, to allow the flexible coupler **20** to pass through the cannulation **11** of sheath **10** (passing through itself and forming a splice **18** within the body of sheath **10**) to form a flexible, tensionable, continuous, adjustable, self-locking, cinching, closed loop **50** around tissue **80**.
(27) Free end **21** of the flexible coupler **20** can be pulled to shrink the construct and the flexible, closed, knotless, continuous, adjustable loops **50**, and to compress the tendon to bone, providing a final repair/construct **101** with increased compression of tissue.
(28) The constructs, systems, and assemblies of the present disclosure may be employed in numerous knotless soft tissue repairs and fixations, for example, fixation of soft tissue to bone.
(29) A soft anchor consists essentially of a flexible tubular member or sheath **10** terminating into a smaller diameter suture **20**. The soft anchor is formed as a one-piece machine taper construct. The suture **20** is a round suture. The suture forms at least one splice within the flexible tubular member **10** and at least one adjustable, knotless, closed, continuous loop **50** around a second tissue **80** to be secured to a first tissue **90**.
(30) Methods of soft tissue repair which do not require tying of knots and allow adjustment of both the tension of the suture and the location of the tissue with respect to the bone, while providing self-locking mechanism, are disclosed. A method of knotless tissue repair comprises inter alia the steps of: securing a flexible cannulated sheath **10** of a one-piece taper construct **100** into a first tissue **90**, the one-piece taper construct **100** being preloaded with a shuttle/pull device **40**; and passing a flexible coupler **20** braided out of the sheath **10** around or through a second tissue **80** to be positioned relative to the first tissue **90** and then through the fixation device **10** by employing the shuttle/pull device, to form an adjustable, knotless, closed, continuous loop **50** around the second tissue **80**. The knotless, closed, adjustable, flexible, continuous loop **50** can have an adjustable perimeter. The flexible coupler can be passed multiple times through the body of the sheath **10**.
(31) As detailed above, when the anchor sheath is inserted, the anchor sheath has at least one repair suture limb (which is fixed to the anchor and manufactured at the same time with the anchor, as a one-piece construct) and also one or more shuttle links. The anchor sheath resides within the bone. The repair suture limb(s) resides on top of the bone. The repair suture is passed around or through the tissue, and then shuttled through the anchor and spliced within the anchor. The steps can be repeated again if additional links are present.
(32) The sheath **10** and braided suture **20** can include any flexible material, for example, multifilament, braided, knitted, woven suture, or including fibers of ultrahigh molecular weight polyethylene (UHMWPE) or the FiberWire® suture (disclosed in U.S. Pat. No. 6,716,234, the

disclosure of which is hereby incorporated by reference in its entirety herein). FiberWire® suture is formed of an advanced, high-strength fiber material, namely ultrahigh molecular weight polyethylene (UHMWPE), sold under the tradenames Spectra (Honeywell) and Dyneema (DSM) fibers, braided with at least one other fiber, natural or synthetic, to form lengths of suture material.
(33) Flexible coupler **20** can be also formed of a stiff material, or combination of stiff and flexible materials, particularly for the regions of the coupler that are passed/spliced through the body of the coupler and depending on whether they are employed with additional fixation devices. In addition, sheath **10** and flexible coupler **20** can be also coated and/or provided in different colors for easy manipulation during the surgical procedure. The knotless constructs and self-locking soft anchors of the present disclosure can be used with any type of flexible material or suture that may be weaved or passed through itself.
(34) Various structural elements of surgical construct **100**, **200**, **300**, **400**, **100***a* may be visually coded, making identification and handling of the sheath and suture legs simpler. Easy identification of suture in situ is advantageous in surgical procedures, particularly during arthroscopic surgeries, endoscopic and laparoscopic procedures.
(35) The surgical constructs of the present disclosure may be employed in endoscopic surgery. The term “endoscopic surgery” refers to surgical procedures within a patient's body through small openings as opposed to conventional open surgery through large incisions. Additionally, surgical constructs as disclosed herein may be utilized in other general surgical and specialty procedures such as soft tissue repairs.
(36) The term “high strength suture” is defined as any elongated flexible member, the choice of material and size being dependent upon the particular application. For the purposes of illustration and without limitation, the term “suture” as used herein may be a cable, filament, thread, wire, fabric, or any other flexible member suitable for tissue fixation in the body.

Claims

1. A soft anchor consisting essentially of a flexible tubular member terminating into a smaller diameter suture, wherein the flexible tubular member is configured to be fixated in bone, and wherein the smaller diameter suture passes through itself and through the flexible tubular member at least once and forms at least one splice within the flexible tubular member.
2. The soft anchor of claim 1, wherein the soft anchor is formed as a one-piece machine taper construct.
3. The soft anchor of claim 1, wherein the smaller diameter suture is a round suture.
4. The soft anchor of claim 1, wherein the smaller diameter suture is a #5 or #2 round suture.
5. The soft anchor of claim 1, wherein the smaller diameter suture is configured to form at least one loop around soft tissue to be attached to bone.
6. The soft anchor of claim 1, wherein the smaller diameter suture passes through the flexible tubular member multiple times.